

EE 446 TinyML - Lab 9

American Sign Language Recognition

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Software Issue Fixed

The intentional issue was the training objective/metric choice. The notebook originally used a regression-style setup with mean squared error and mean absolute error for a multi-class gesture classification task. Because the model output is a softmax over gesture classes and the labels are one-hot encoded, this was not the correct loss or metric for the problem.

Correction made: I changed the model compile step to use categorical cross-entropy loss and categorical accuracy. This makes the training objective match the classification problem.

```
model.compile(  
    optimizer='rmsprop',  
    loss='categorical_crossentropy',  
    metrics=['categorical_accuracy']  
)
```

Notebook / Export Consistency Changes

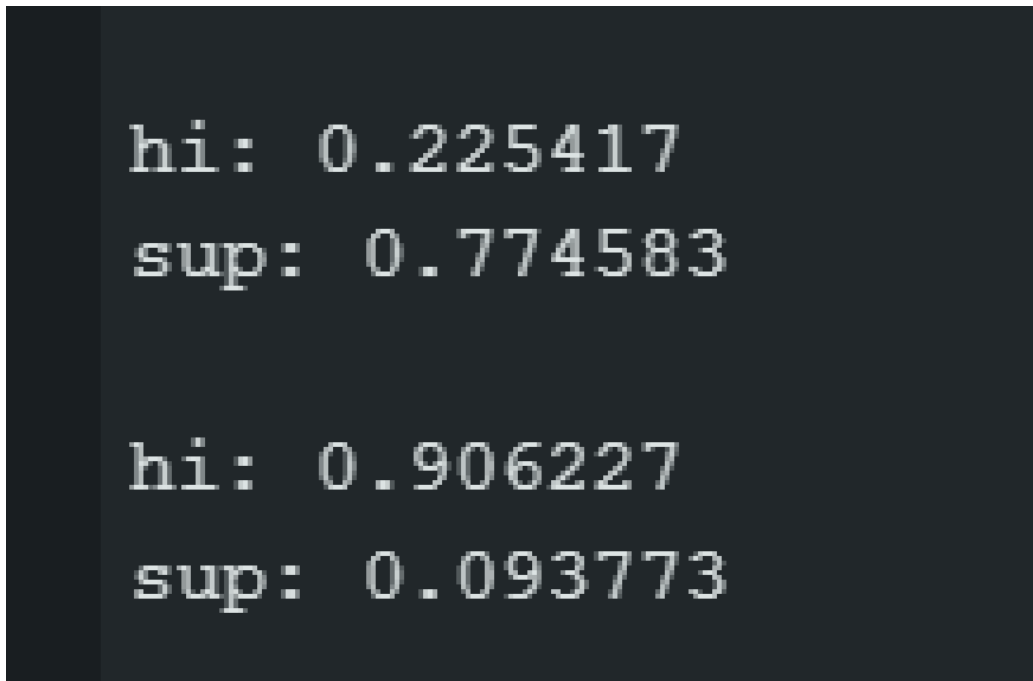
- Updated downstream metric references from MAE-style plots to categorical accuracy plots.
- Updated plot titles, axis labels, and history keys so they match the corrected metric name.
- Kept the model output as a softmax classifier over the two gesture labels: hi and sup.
- Exported the trained TensorFlow Lite model into model.h so it could be included by the Arduino sketch.
- Used the Arduino sketch to read IMU data, run inference, and print the predicted gesture confidence scores in the Serial Monitor.

Hardware Test Results

The Arduino Serial Monitor successfully identified both required gestures. For the first test, the model gave the highest confidence to sup. For the second test, the model gave the highest confidence to hi. These results show that the deployed TinyML model is running on the board and producing the expected gesture predictions.

Test	hi confidence	sup confidence	Predicted gesture
Gesture test 1	0.225417	0.774583	sup
Gesture test 2	0.906227	0.093773	hi

Arduino Serial Monitor Output



```
hi: 0.225417  
sup: 0.774583  
  
hi: 0.906227  
sup: 0.093773
```

The image shows a screenshot of an Arduino Serial Monitor. The background is dark, and the text is light gray. It displays two sets of output. The first set shows 'hi: 0.225417' and 'sup: 0.774583'. After a blank line, the second set shows 'hi: 0.906227' and 'sup: 0.093773'. The text is in a monospaced font.

Figure 1. Serial Monitor output showing both ASL gesture classes identified by the deployed model.

Conclusion

After correcting the software issue and keeping the notebook outputs consistent with the classification setup, the model was exported and deployed successfully. The hardware test confirmed that the Arduino implementation can distinguish between the two trained gestures, hi and sup, using the Serial Monitor confidence scores.